Gaze-Weighted Winning Designs, Per Myopia Level

Visualized to Scale, and Fabrication Decision

IYPT Pinhole Sunglasses — Array Geometry Sub-Study

August 17, 2026

Contents

1 Summary Table **2**

2 M = 0 D and Global Scope **3**

3 M = 1 D **3**

4 M = 2–3 D **4**

5 M = 4–5 D **5**

6 M = 6 D, M = 8 D **5**

7 Should Fabrication Start Now? **6**

7.1 Which Design to Build First 6

7.2 What to Measure 6

1 Summary Table

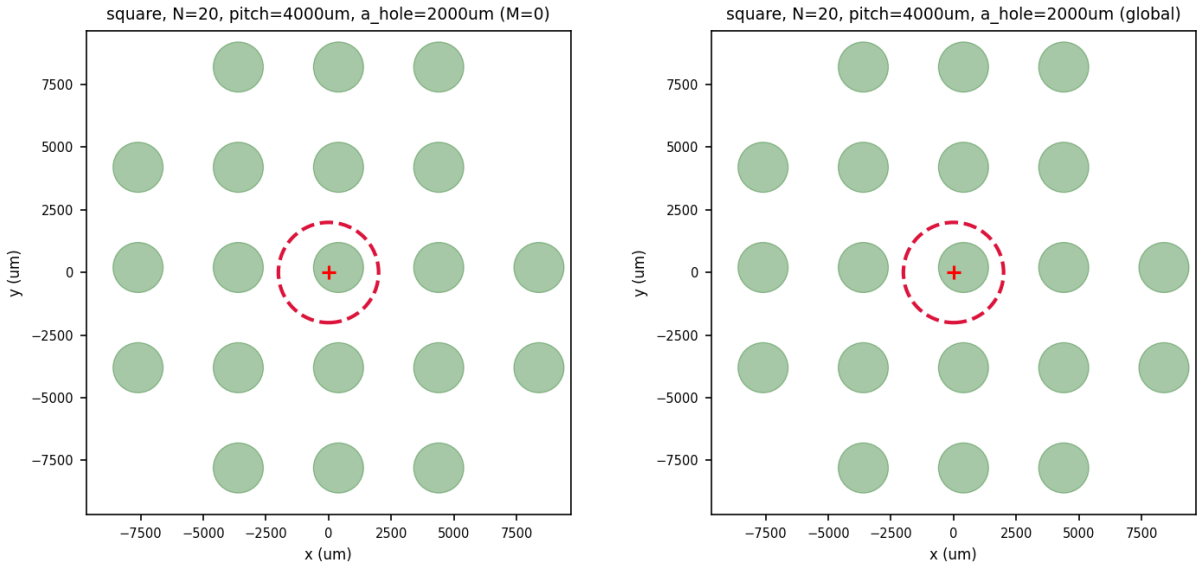
All values below are gaze-weighted (integrated over the measured eccentricity distribution, azimuthally sampled), not worst-case. Green circles are pinholes to true scale; the dashed red circle is the 4 mm pupil at $\varphi = 0$. All designs use the sparsest tested pitch (4000 μm) — confirmed to be a genuine pattern, not a display choice.

M (D)	Family	Spacing/N	Ghost %	Blind %
0	square	pitch=4000, N=20	0.00	0.00
1	hexagonal	pitch=4000, N=20	0.00	0.00
2–3	hexagonal	pitch=4000, N=20	0.00	0.00
4–5	hexagonal	pitch=4000, N=20	0.00	0.00
6	hexagonal	pitch=4000, N=100	0.00	0.12
8	hexagonal	pitch=4000, N=50	0.00	0.33
global	square	pitch=4000, N=20	0.00	0.00

Note

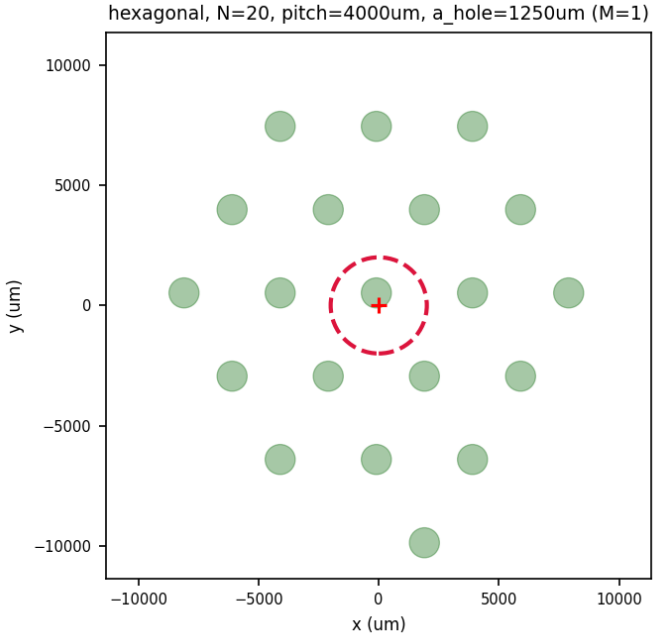
Every design here also carries $+\infty$ worst-case ghost and vignetting margin under this project's primary (stricter) safety gate — these are not a weaker alternative set, they are designs that pass both the conservative worst-case standard AND achieve genuine 0% ghosting under realistic, time-weighted gaze. A large tied plateau of designs reaches exactly 0.00%/0.00% at several M; the specific family/N shown is one representative of that tie, not a uniquely sharper winner.

2 M = 0 D and Global Scope



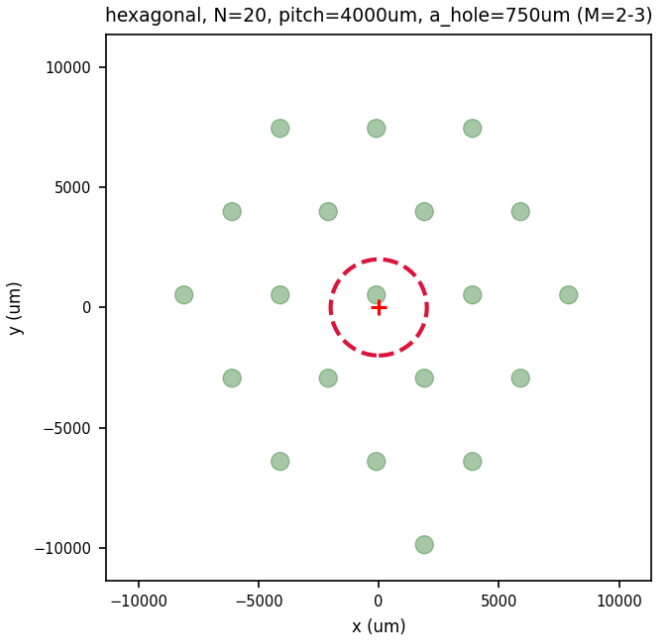
`square`, pitch=4000 μm, N=20, a_hole=2000 μm — identical geometry serves both M=0 and the global (worst-case-across-M) scope. 0.00%/0.00% ghost/blind.

3 M = 1 D



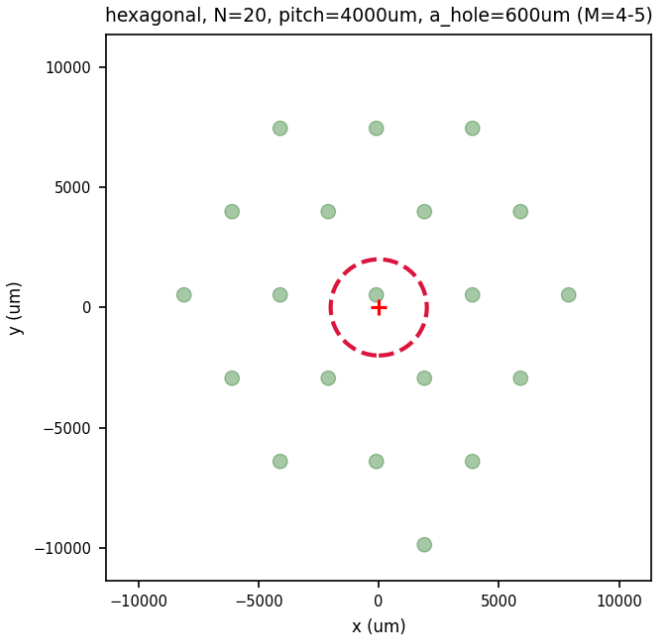
`hexagonal`, pitch=4000 μm, N=20, a_hole=1250 μm. 0.00%/0.00%.

4 M = 2–3 D



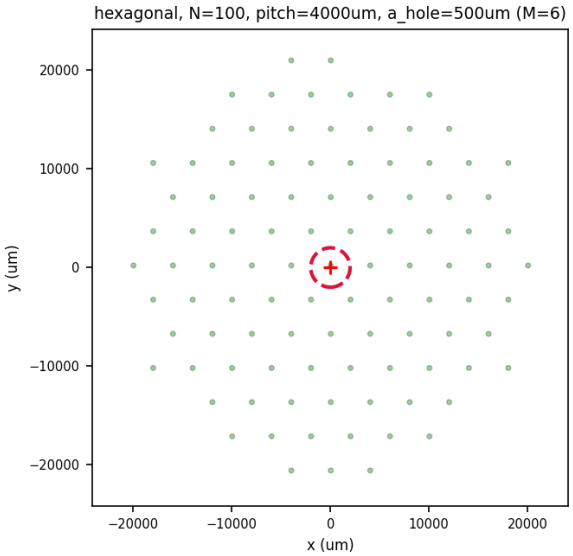
`hexagonal`, pitch=4000 μm, N=20, a_hole=750 μm. 0.00%/0.00%.

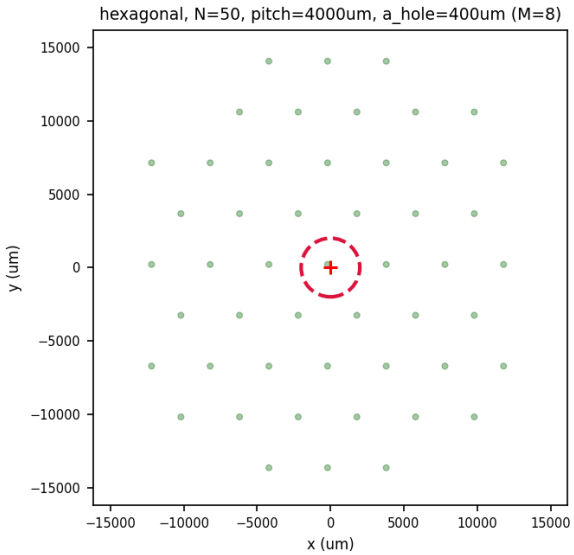
5 M = 4–5 D



`hexagonal`, pitch=4000 μm, N=20, a_hole=600 μm. 0.00%/0.00%.

6 M = 6 D, M = 8 D





`hexagonal`, N=100 (M=6, left) and N=50 (M=8, right), same pitch and family. These are the only two designs in the set with nonzero blind rate (0.12% and 0.33%) — still ghost-free, and both blind percentages are smaller than the smallest nonzero value in the worst-case-optimal table.

7 Should Fabrication Start Now?

Fabrication Guidance

Yes. The theoretical model has been through defect-finding, margin-aware re-verification, hand-cross-checking, cross-validation against a competing published approach, seed-robustness testing, and now two independently-computed safety standards (worst-case and gaze-weighted) that agree on which designs are safe. Further theoretical iteration has reached diminishing returns; a physical prototype is the highest-value next step.

7.1 Which Design to Build First

The gaze-weighted result changes the practical fabrication plan for the better: **one family (hexagonal), one pitch (4000 μm), covering M=1 through M=8 with only the hole diameter changing between inserts** (1250, 750, 600, 500, 400 μm). This is simpler to fabricate than the mixed-family worst-case-optimal set (`ring_offset_concentric` + `hexagonal` + `square` across different M ranges), and every design in it achieves genuine 0.00% ghosting under realistic use, not just a structural guarantee at one gaze angle.

Recommended build order:

1. **Hexagonal, pitch=4000 μm, N=20, a_hole=400 μm** (M=8 insert) — smallest hole, test cutting-edge quality first, per the established fabrication discipline.
2. **Hexagonal, pitch=4000 μm, N=20, a_hole=750 μm** (M=2–3 insert) — mid-range, a second data point on the same base frame.
3. **Square, pitch=4000 μm, N=20, a_hole=2000 μm** (M=0/global) — confirms the model at the opposite end of the aperture range, different family for a cross-check.

7.2 What to Measure

Per the project's established phone-camera afocal protocol: blur at each design's own M setting, and ghosting specifically at gaze angles beyond 17° (the 95%-of-time boundary) — this is the one region where the worst-case and gaze-weighted models could plausibly diverge from reality in different ways, and it is the most informative test of which safety standard the real optics actually respect.